

03

Matrix Multiplication

Matrix Multiplication

Inner product, outer product, column-vector combination, row-vector combination ...

3.1 Matrix Multiplication: The First Perspective



Core skills you'll master in this section:

- ▶ Aligning rows and columns to understand shape compatibility for multiplication: when multiplying $\mathbf{AB}$, place $\mathbf{A}$'s rows aligned on the left, and $\mathbf{B}$'s columns aligned on top — this lets you directly judge whether $\mathbf{AB}$ is valid and determine the shape of the result.
- ▶ The inner-product perspective for computing elements: in matrix multiplication, each element is the inner product of a row of $\mathbf{A}$ and a column of $\mathbf{B}$.
- ▶ A vector inner product can be converted into matrix multiplication: a 1×1 matrix is equivalent to a scalar.
- ▶ A vector outer product generates a matrix: constructing a rank-one matrix (covered in detail in the next section).

The previous chapter already introduced matrix multiplication; this chapter is devoted to understanding matrix multiplication from different perspectives.

The first perspective on matrix multiplication is an intuitive way to understand matrix multiplication: by aligning matrix $\mathbf{A}$ and matrix $\mathbf{B}$ in a specific way, it helps us better understand the shape requirements of matrix multiplication, as well as how each element is computed.

Row and Column Alignment

As shown in Figure 1, matrix $\mathbf{A}$ is an $m \times p$ matrix, and matrix $\mathbf{B}$ is a $p \times n$ matrix.

Place matrix $\mathbf{A}$ to the left of the result matrix $\mathbf{C}$, row-aligned; at the same time, place matrix $\mathbf{B}$ on top of $\mathbf{C}$, column-aligned. This alignment lets us clearly see the shape rule for matrix multiplication.

Placing $\mathbf{A}$ to the left of $\mathbf{C}$ means that $\mathbf{A}$'s row count m determines $\mathbf{C}$'s row count m .

Placing $\mathbf{B}$ on top of $\mathbf{C}$ means that $\mathbf{B}$'s column count n determines $\mathbf{C}$'s column count n .

Therefore, in the matrix multiplication $A @ B = C$ gives the result matrix C with shape $m \times n$.

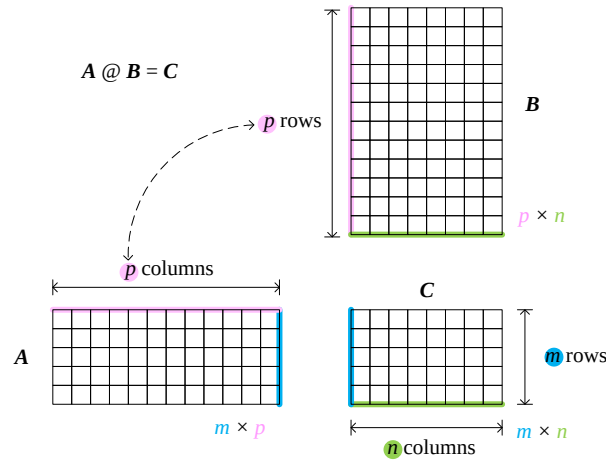


Figure 1. A Is Placed to the Left of AB , Row-Aligned; B Is Placed on Top of AB , Column-Aligned

This alignment also reveals the core rule of matrix multiplication: A 's number of columns must equal B 's number of rows, otherwise they cannot be aligned.

As shown in Figure 2, after aligning, C 's every element c_{ij} is A 's row i and B 's column j — the result of multiplying and summing the corresponding entries.

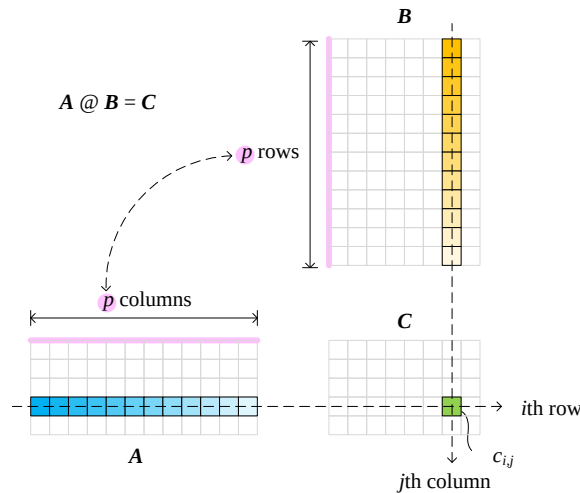


Figure 2. Computing AB 's Row i , Column j Element: A Is Placed to the Left of AB , Row-Aligned; B Is Placed on Top of AB , Column-Aligned

This spatially-aligned perspective not only helps us understand the shape of matrix multiplication, but also lets us more intuitively feel how matrix multiplication integrates the information from two matrices in a specific way.

Computing Each Element

As mentioned earlier in this book, matrix $\mathbf{A}$ can be written as a set of row vectors, each row vector denoted $\mathbf{a}^{(i)}$; matrix $\mathbf{B}$ can be written as a set of column vectors, each column vector denoted $\mathbf{b}_j$. We can write the matrix multiplication $\mathbf{AB}$ as several matrix multiplications (the inner-product perspective).

As shown in Figure 3, let's write the matrix product $\mathbf{C} = \mathbf{AB}$'s row- i , column- j element c_{ij} as a single matrix multiplication

$$c_{i,j} = \mathbf{a}^{(i)} @ \mathbf{b}_j$$

Of course, we can also compute c_{ij} using a vector inner product, that is

$$c_{i,j} = \mathbf{a}^{(i)\top} \cdot \mathbf{b}_j$$

Therefore, this first perspective on matrix multiplication is also called the inner-product expansion.

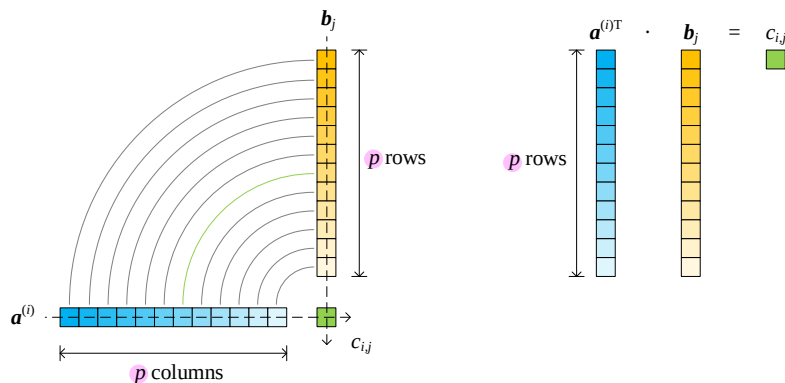


Figure 3. The Inner-Product-Expansion Perspective on the Matrix Product $\mathbf{a}^{(i)} @ \mathbf{b}_j = c_{ij}$

Inner Product

also tells us that the inner product of two column vectors of equal dimension can also be written in the form of matrix multiplication.

For example, as shown in Figure 4, for the n -dimensional vectors $\mathbf{a}$ and $\mathbf{b}$, their inner product is equivalent to

$$\mathbf{a} \cdot \mathbf{b} = \langle \mathbf{a}, \mathbf{b} \rangle = \mathbf{a}^\top \mathbf{b} = \mathbf{b}^\top \mathbf{a}$$

This is exactly the vector inner product (inner product) introduced earlier in this book.

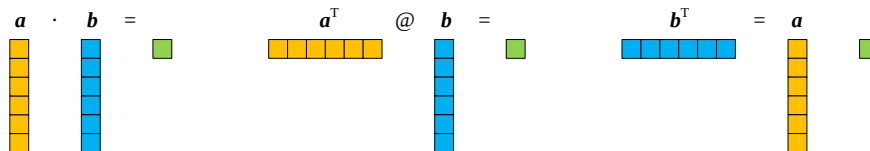


Figure 4. A Vector Inner Product Can Also Be Written in the Form of Matrix Multiplication

For example, given the vectors of equal dimension

$$\mathbf{a} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

Use the matrix multiplication $\mathbf{a}^T \mathbf{b}$ to compute their inner product

$$\mathbf{a}^T @ \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}^T @ \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = [1 \ 2 \ 3] @ \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = 32$$

Use the matrix multiplication $\mathbf{b}^T \mathbf{a}$ to compute their inner product

$$\mathbf{b}^T @ \mathbf{a} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}^T @ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = [4 \ 5 \ 6] @ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 32$$

Looking at the matrix shapes, $(1 \times 3) @ (3 \times 1)$ — the (3) sandwiched in the middle gets “canceled out,” so the result of the matrix multiplication has shape (1×1) . As mentioned earlier in this book, a 1×1 matrix can be regarded as a scalar.

As shown in Figure 5, column vector $\mathbf{a}$'s length (magnitude, L^2 norm, modulus, Euclidean norm) squared can be obtained either through an inner product or via matrix multiplication, that is

$$\|\mathbf{a}\|^2 = \|\mathbf{a}\|_2^2 = \mathbf{a} \cdot \mathbf{a} = \langle \mathbf{a}, \mathbf{a} \rangle = \mathbf{a}^T \mathbf{a}$$

Seen this way, $\mathbf{a}$'s length (magnitude, L^2 norm, modulus, Euclidean norm) is simply the square root of the value in the equation above

$$\|\mathbf{a}\| = \|\mathbf{a}\|_2 = \sqrt{\mathbf{a} \cdot \mathbf{a}} = \sqrt{\langle \mathbf{a}, \mathbf{a} \rangle} = \sqrt{\mathbf{a}^T \mathbf{a}}$$

Figure 5. The Square of a Vector's Length Can Also Be Written in the Form of Matrix Multiplication

Use matrix multiplication to compute the length of vector $\mathbf{a}$ (magnitude, L^2 norm, modulus, Euclidean norm)

$$\|\mathbf{a}\| = \|\mathbf{a}\|_2 = \sqrt{\mathbf{a}^T \mathbf{a}} = \sqrt{[1 \ 2 \ 3] @ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}} = \sqrt{14}$$

Outer Product

The counterpart to the inner product is the outer product (outer product).

If the inner product of two vectors results in a scalar, then the outer product of two vectors results in a matrix — shown specifically in Figure 6. This special form of matrix multiplication will help us understand the “second perspective on matrix multiplication” in the next section.

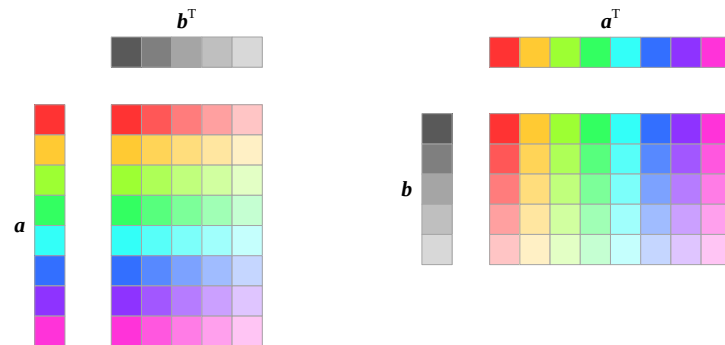


Figure 6. Two Outer Products of Vectors $\mathbf{a}$ and $\mathbf{b}$

For example, given two column vectors (not required to have equal dimension)

$$\mathbf{a} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Compute the first outer product $\mathbf{a} @ \mathbf{b}^T$

$$\mathbf{a} @ \mathbf{b}^T = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{bmatrix} @ \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \\ 4 & 8 & 12 \\ 5 & 10 & 15 \end{bmatrix}$$

Looking at the matrix shapes, $(5 \times 1) @ (1 \times 3)$ — the (1) sandwiched in the middle gets “canceled out,” so the result of the matrix multiplication has shape (5×3) .

Now compute the second outer product $\mathbf{b} @ \mathbf{a}^T$

$$\mathbf{b} @ \mathbf{a}^T = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} @ \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \\ 4 & 8 & 12 \\ 5 & 10 & 15 \end{bmatrix}$$

Looking at the matrix shapes, $(3 \times 1) @ (1 \times 5)$ — the (1) sandwiched in the middle gets “canceled out,” so the result of the matrix multiplication has shape (3×5) .

The First Example

Let's return to a matrix-multiplication example used earlier in this book

$$\mathbf{A} @ \mathbf{B} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} @ \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} 14 & 32 \\ 32 & 77 \end{bmatrix}$$

$\mathbf{A}$ is written as row vectors, $\mathbf{B}$ is written as column vectors, then $\mathbf{A} @ \mathbf{B}$ can be written as

$$\begin{aligned} \mathbf{A} @ \mathbf{B} &= \begin{bmatrix} \mathbf{a}^{(1)} \\ \mathbf{a}^{(2)} \end{bmatrix} [\mathbf{b}_1 \quad \mathbf{b}_2] = \begin{bmatrix} \mathbf{a}^{(1)} \mathbf{b}_1 & \mathbf{a}^{(1)} \mathbf{b}_2 \\ \mathbf{a}^{(2)} \mathbf{b}_1 & \mathbf{a}^{(2)} \mathbf{b}_2 \end{bmatrix} \\ &= \begin{bmatrix} [1 & 2 & 3] \\ [4 & 5 & 6] \end{bmatrix} @ \begin{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} & \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} [1 & 2 & 3] @ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} & [1 & 2 & 3] @ \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} \\ [4 & 5 & 6] @ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} & [4 & 5 & 6] @ \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} \end{bmatrix} \\ &= \begin{bmatrix} 14 & 32 \\ 32 & 77 \end{bmatrix} \end{aligned}$$

As shown in Figure 7, using the first perspective on matrix multiplication, you may have already noticed that the equation above contains four matrix-multiplication operations, namely

$$\begin{cases} c_{1,1} = \mathbf{a}^{(1)} @ \mathbf{b}_1 \\ c_{1,2} = \mathbf{a}^{(1)} @ \mathbf{b}_2 \\ c_{2,1} = \mathbf{a}^{(2)} @ \mathbf{b}_1 \\ c_{2,2} = \mathbf{a}^{(2)} @ \mathbf{b}_2 \end{cases}$$

For example, $\mathbf{a}^{(1)} @ \mathbf{b}_1$ is one matrix multiplication; a 1×3 row vector $\mathbf{a}^{(1)}$ and a 3×1 column vector $\mathbf{b}_1$ multiply together to give a 1×1 matrix, which is equivalent to a scalar.

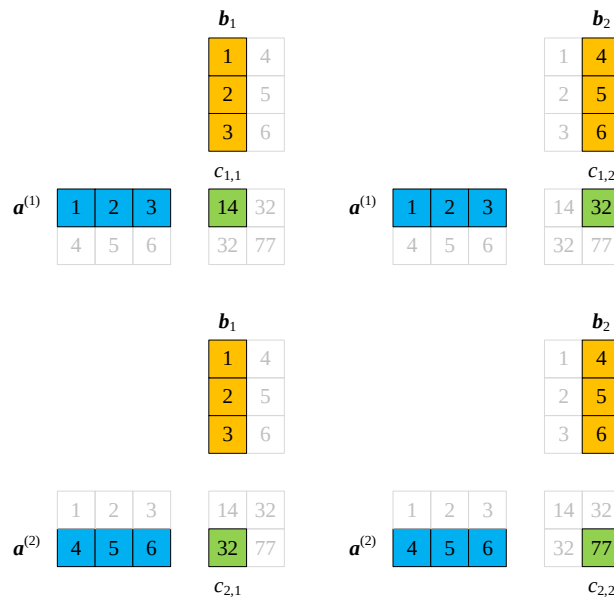


Figure 7. $\mathbf{A}$ Is Placed to the Left of $\mathbf{AB}$, Row-Aligned; $\mathbf{B}$ Is Placed on Top of $\mathbf{AB}$, Column-Aligned

The Second Example

Let's look at a second example: compute the matrix product $\mathbf{B} @ \mathbf{A}$

$$\mathbf{D} = \mathbf{B} @ \mathbf{A} = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} @ \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 17 & 22 & 27 \\ 22 & 29 & 36 \\ 27 & 36 & 45 \end{bmatrix}$$

$\mathbf{B}$ is written as row vectors, $\mathbf{A}$ is written as column vectors, then $\mathbf{B} @ \mathbf{A}$ can be written as

$$\begin{aligned} \mathbf{B} @ \mathbf{A} &= \begin{bmatrix} \mathbf{b}^{(1)} \\ \mathbf{b}^{(2)} \\ \mathbf{b}^{(3)} \end{bmatrix} \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 \end{bmatrix} = \begin{bmatrix} \mathbf{b}^{(1)} \mathbf{a}_1 & \mathbf{b}^{(1)} \mathbf{a}_2 & \mathbf{b}^{(1)} \mathbf{a}_3 \\ \mathbf{b}^{(2)} \mathbf{a}_1 & \mathbf{b}^{(2)} \mathbf{a}_2 & \mathbf{b}^{(2)} \mathbf{a}_3 \\ \mathbf{b}^{(3)} \mathbf{a}_1 & \mathbf{b}^{(3)} \mathbf{a}_2 & \mathbf{b}^{(3)} \mathbf{a}_3 \end{bmatrix} \\ &= \begin{bmatrix} [1 & 4] \\ [2 & 5] \\ [3 & 6] \end{bmatrix} @ \begin{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} & \begin{bmatrix} 2 \\ 5 \end{bmatrix} & \begin{bmatrix} 3 \\ 6 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} [1 & 4] @ \begin{bmatrix} 1 \\ 4 \end{bmatrix} & [1 & 4] @ \begin{bmatrix} 2 \\ 5 \end{bmatrix} & [1 & 4] @ \begin{bmatrix} 3 \\ 6 \end{bmatrix} \\ [2 & 5] @ \begin{bmatrix} 1 \\ 4 \end{bmatrix} & [2 & 5] @ \begin{bmatrix} 2 \\ 5 \end{bmatrix} & [2 & 5] @ \begin{bmatrix} 3 \\ 6 \end{bmatrix} \\ [3 & 6] @ \begin{bmatrix} 1 \\ 4 \end{bmatrix} & [3 & 6] @ \begin{bmatrix} 2 \\ 5 \end{bmatrix} & [3 & 6] @ \begin{bmatrix} 3 \\ 6 \end{bmatrix} \end{bmatrix} \\ &= \begin{bmatrix} 17 & 22 & 27 \\ 22 & 29 & 36 \\ 27 & 36 & 45 \end{bmatrix} \end{aligned}$$

Figure 8 shows how to use the first perspective on matrix multiplication to understand the computation rule for each element of $\mathbf{B} @ \mathbf{A}$.

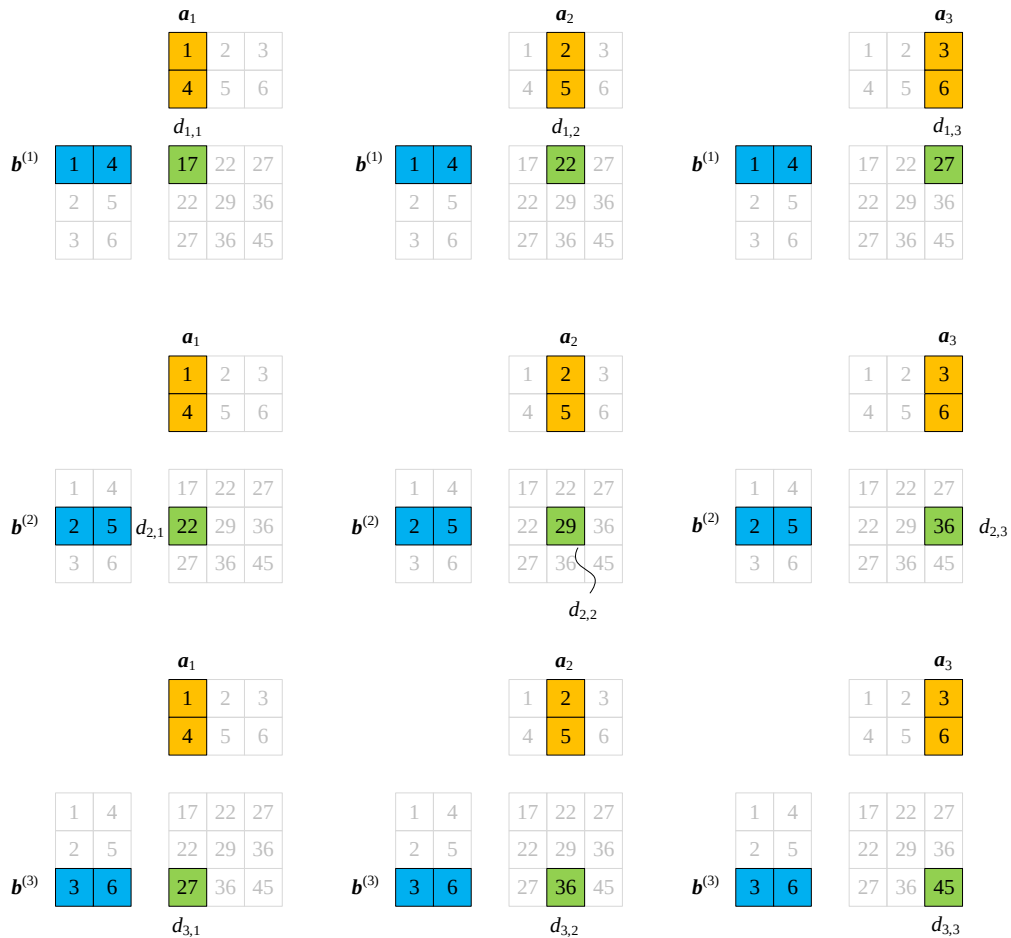


Figure 8. Computing Each Element: B Is Placed to the Left of BA , Row-Aligned; A Is Placed on Top of BA , Column-Aligned



LA_03_01_01.ipynb computes the matrix multiplication above using the first perspective on matrix multiplication.

A Custom Python Function to Compute Matrix Multiplication: The Inner-Product Perspective

Code 1 describes a custom Python function to compute matrix multiplication; this code uses the inner-product perspective introduced in this section. Let's go over the key statements below. Note that statements already covered in the previous chapter are skipped here – we recommend reviewing them.

- a** Using `def` defines a custom function named `matrix_multiplication_inner`, which takes two inputs: matrix A and matrix B .
- b** Starts a loop. This means “starting from row 0, up through row $m-1$ ”, taking one row of matrix A to process each time. i is the current row number.
- c** This is the second, nested loop. It means “process every column, one at a time”, from column 0 to column $n-1$. j is the current column number. Here, the `numpy.dot()` function computes the “inner product” of `row_i_A` and `col_j_B`. The inner product is the result of multiplying and summing a row and a column together – this is the value of a single element in matrix multiplication.

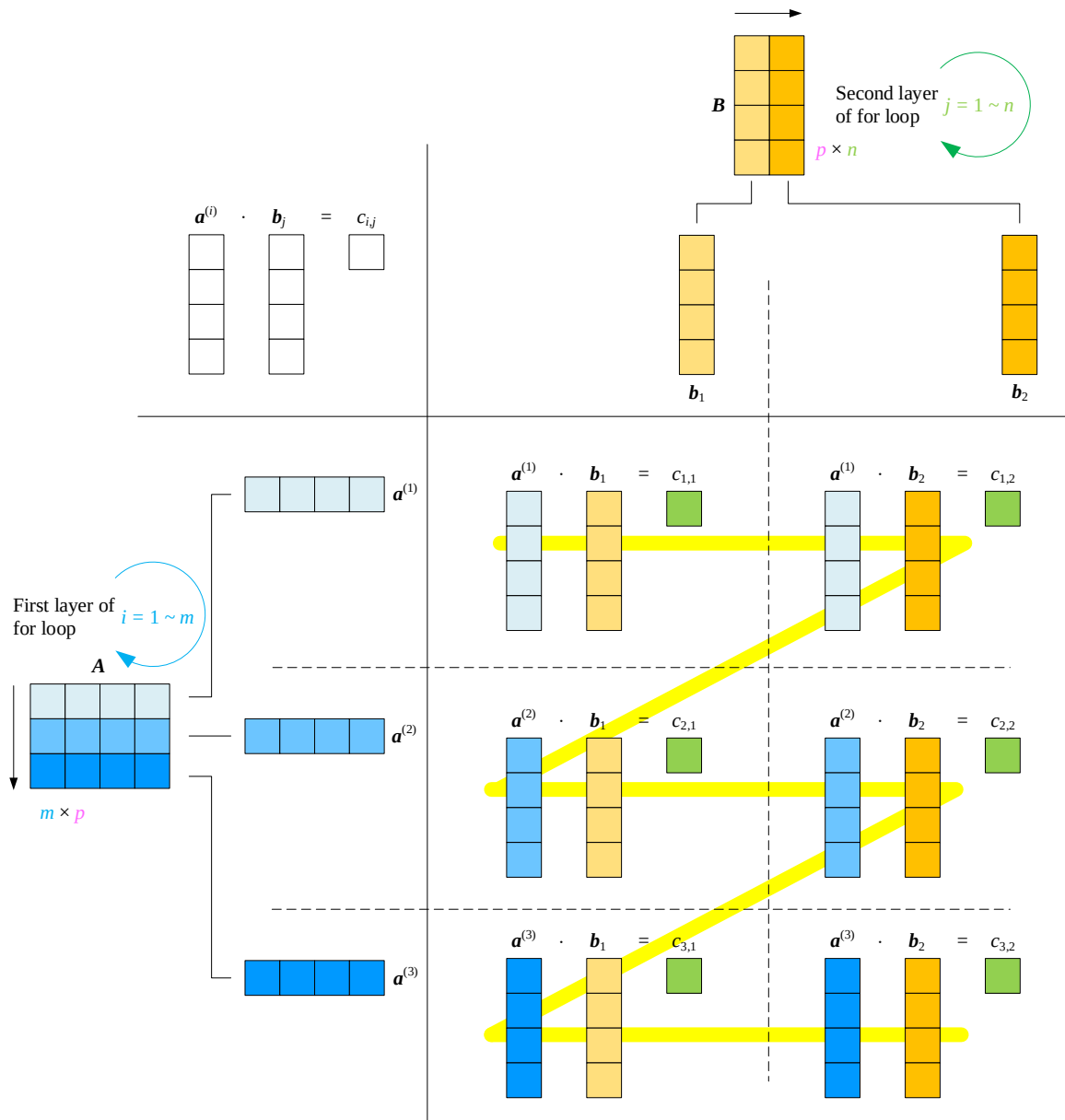
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Code and PDF file downloads: <https://github.com/Visualize-ML>

The companion video lectures for this book are all published on Bilibili — 生姜 DrGinger: <https://space.bilibili.com/513194466>

Comments and corrections are welcome. The book's dedicated email address: jiang.visualize.ml@gmail.com

Figure 9. The Rule for the Matrix Multiplication AB , the Inner-Product Perspective

Code 1. A Custom Python Function to Compute Matrix Multiplication, the Inner-Product Perspective | 
 LA_03_01_02.ipynb

```

## 初始化
import numpy as np

## 自定义函数，矩阵乘法第一视角
def matrix_multiplication_inner(A, B):

    # 获取矩阵 A 和 B 的形状
    m, p_A = A.shape
    p_B, n = B.shape

    # 检测矩阵形状是否符合矩阵乘法规则
    if p_A != p_B:
        raise ValueError('Dimensions do not match')

    # 初始化结果矩阵 C，形状 (m, n)，初始值设为 0
    C = np.zeros((m, n))

    # 计算每个元素
    for i in range(m):
        for j in range(n):
            row_i_A = A[i, :]          # A 的第 i 行（行向量）
            col_j_B = B[:, j]          # B 的第 j 列（列向量）
            C[i, j] = np.dot(row_i_A, col_j_B) # 计算内积

    return C

## 矩阵乘法
A = np.array([[1, 2, 3],
              [4, 5, 6]])
B = A.T

## 矩阵乘法
matrix_multiplication_inner(A, B)
matrix_multiplication_inner(B, A)

```



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Given the column vector $\mathbf{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, use the first-perspective method to place and compute $\mathbf{a}^T @ \mathbf{a}$, $\mathbf{a} @ \mathbf{a}^T$.

Q2. Given the column vectors $\mathbf{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, use the first-perspective method to place and compute $\mathbf{a} @ \mathbf{b}^T$, $\mathbf{b} @ \mathbf{a}^T$.

Q3. Given $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, use the first-perspective method to place and compute $A @ b$, $b^T @ A^T$.

Q4. Given $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$, use the first-perspective method to place and compute $A @ A^T$, $A^T @ A$.

Q5. Modify Code 1, swapping the inner and outer for loops.